Twinning

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Learning objectives

-Define dizygotic and monozygotic twinning.

-Explain how the timing of monozygotic embryo division determines chorionic and amniotic cavities.

-Recognize the main placental and membrane arrangements in twin pregnancies.

-Describe important complications, especially those related to monochorionic twins.

What is twinning?

Twinning = development of two offspring from a single pregnancy.

Dizygotic (DZ) twins: two ova are fertilized by two sperm → two zygotes.

Monozygotic (MZ) twins: one fertilized ovum → zygote → gives rise to two embryos.

Chorion = number of chorions/placentas.

Amnion = number of amniotic sacs.

Dizygotic (fraternal) twins

Mechanism: two separate oocytes are ovulated and fertilized by two different sperm.

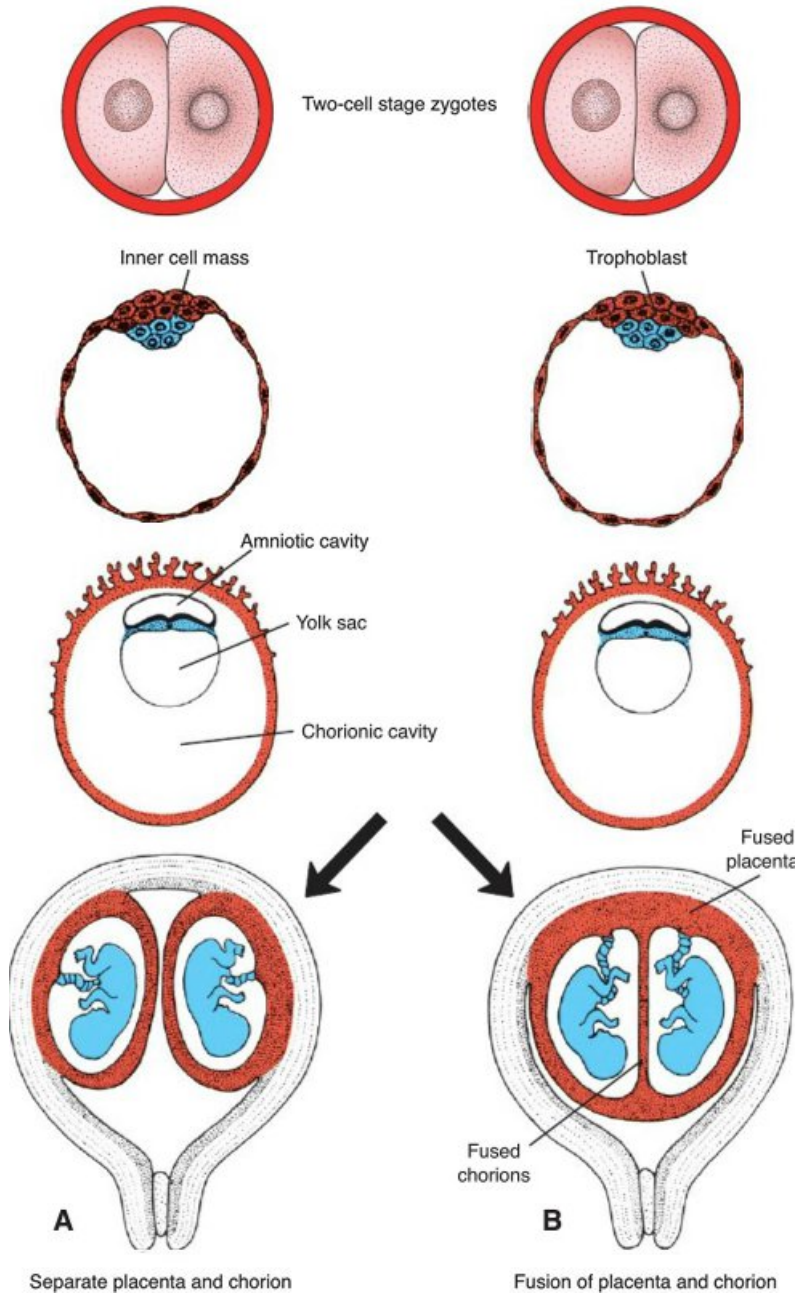
Genetically similar to ordinary siblings; different sex is possible.

Each embryo develops its own chorion and amnion.

Typical arrangement: Dichorionic–Diamniotic (DCDA).

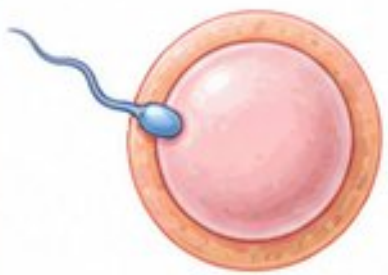
Two placentas may remain separate or fuse together.

Incidence is influenced by maternal age, parity, family history, and assisted reproduction.


Two-cell stage zygotes
Inner cell mass
Trophoblast
Amniotic cavity
Yolk sac
Chorionic cavity
Fused placenta
Fused chorions
A
B
Separate placenta and chorion
Fusion of placenta and chorion

1. Stages of Monozygotic (Identical) Twinning

1 Fertilization



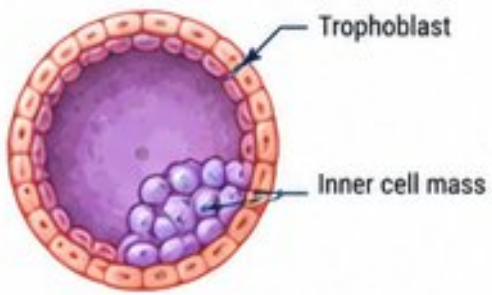
One ovum is fertilized by one sperm (one zygote).

2 Early cleavage (Day 1–3)



The zygote undergoes cleavage to form a morula (8–16 cells).

3 Blastocyst stage (Day 4–5)



The inner cell mass (ICM) can split, forming two embryos.

4 Division of ICM (early monozygotic twins)



Two separate embryos develop, each with its own amnion and chorion (depending on timing).

5 Two embryos implant



Each embryo implants and develops, leading to two fetuses (identical).

Monozygotic twinning: timing of embryonic division

Days 1–3: division before formation of the chorion → Dichorionic–Diamniotic.

Days 4–8: division after chorion formation but before amnion formation → Monochorionic–Diamniotic.

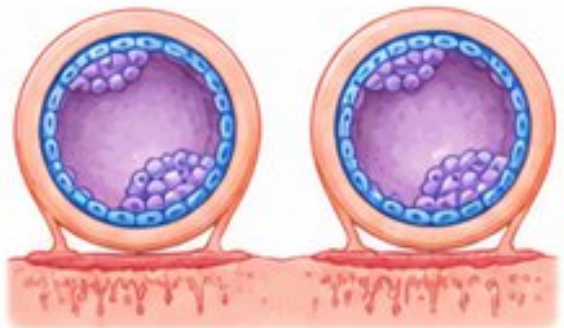
Days 9–13: division after amnion formation → Monochorionic–Monoamniotic.

After day 13: incomplete separation may occur → Conjoined twins.

Later separation produces more shared structures.

2. Membrane Formation and Placental Outcomes

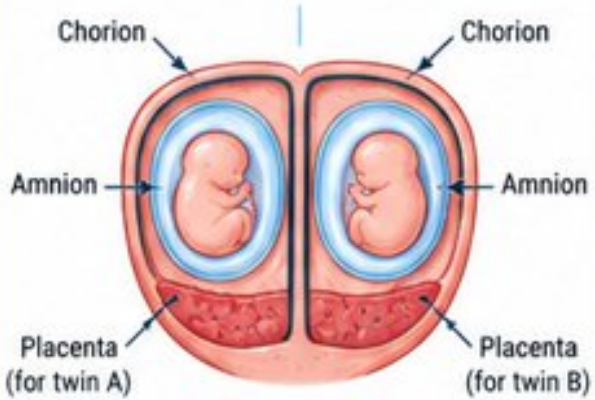
A. Dizygotic (Fraternal) Twins



- Two separate ova, two sperms
- Always dichorionic–diamniotic (DCDA)
- Separate placentas (may be fused)

Timing: Any time (after fertilization)

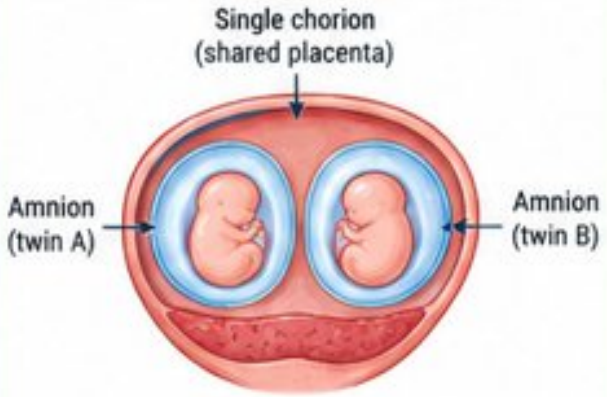
B. Monozygotic Twins – DCDA (division ≤ Day 3)



- Division before day 3
- Two chorions, two amnions
- Often separate placentas

Timing: ≤ Day 3

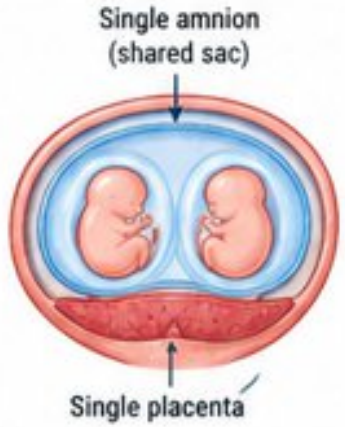
C. Monozygotic Twins – MCDA (division Day 4–8)



- Division Day 4–8
- One chorion, two amnions
- Single shared placenta

Timing: Day 4 – 8

D. Monozygotic Twins – MCMA (division ≥ Day 9)



- Division ≥ Day 9
- One chorion, one amnion
- Single shared placenta

Timing: ≥ Day 9

MZ twins: timing and membranes

Approx. timing	Embryologic event	Chorionicity	Amnionicity	Typical result
Days 1–3	Early blastomere/inner cell mass separation	Dichorionic	Diamniotic	DCDA
Days 4–8	Separation after chorion develops	Monochorionic	Diamniotic	MCDA
Days 8–13	Separation after amnion develops	Monochorionic	Monoamniotic	MCMA
> Day 13	Incomplete separation	Monochorionic	Monoamniotic	Conjoined twins

High-yield rule: the later the embryo separates, the more fetal membranes are shared.

3. Summary of Membranes and Placentation

Type	Chorions	Amnions	Placentas	Timing of division
Dizygotic (fraternal)	2	2	2 (or fused)	Any time
Monozygotic – DCDA	2	2	2 (or fused)	≤ Day 3
Monozygotic – MCDA	1	2	1 (shared)	Day 4 – 8
Monozygotic – MCMA	1	1	1 (shared)	≥ Day 9

4. Key Visual Comparison

DCDA



Two chorions
Two amnions
Two placentas

MCDA



One chorions
Two amnions
One placenta

MCMA



One chorion
One amnion
One placenta

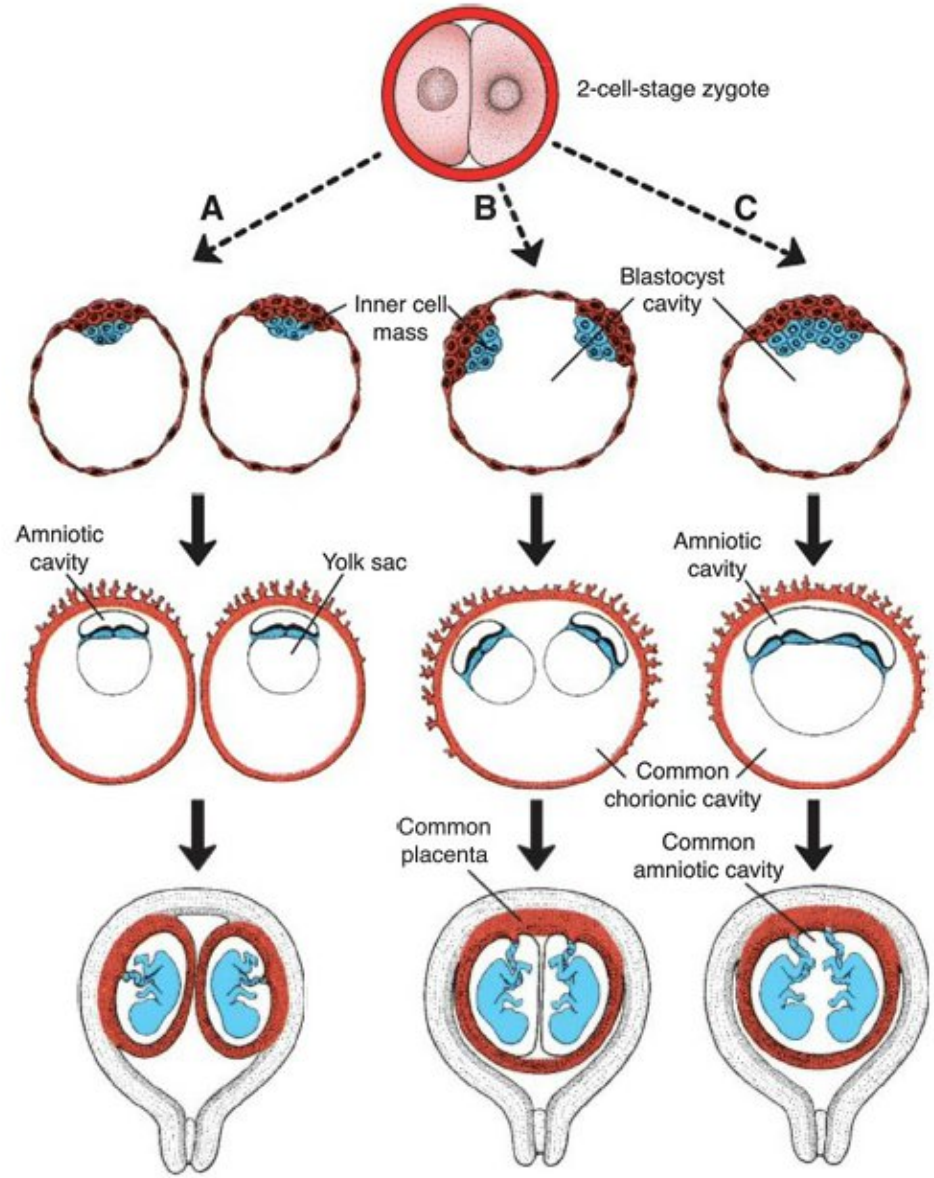


FIGURE 8.19 Possible relations of fetal membranes in monozygotic twins. **A.** Splitting occurs at the two-cell stage, and each embryo has its own placenta, amniotic cavity, and chorionic cavity. **B.** Splitting of the inner cell mass into two completely separated groups. The two embryos have a common placenta and a common chorionic sac but separate amniotic cavities. **C.** Splitting of the inner cell mass at a late stage of development. The embryos have a common placenta, a common amniotic cavity, and a common chorionic cavity.

Why chorion matters clinically

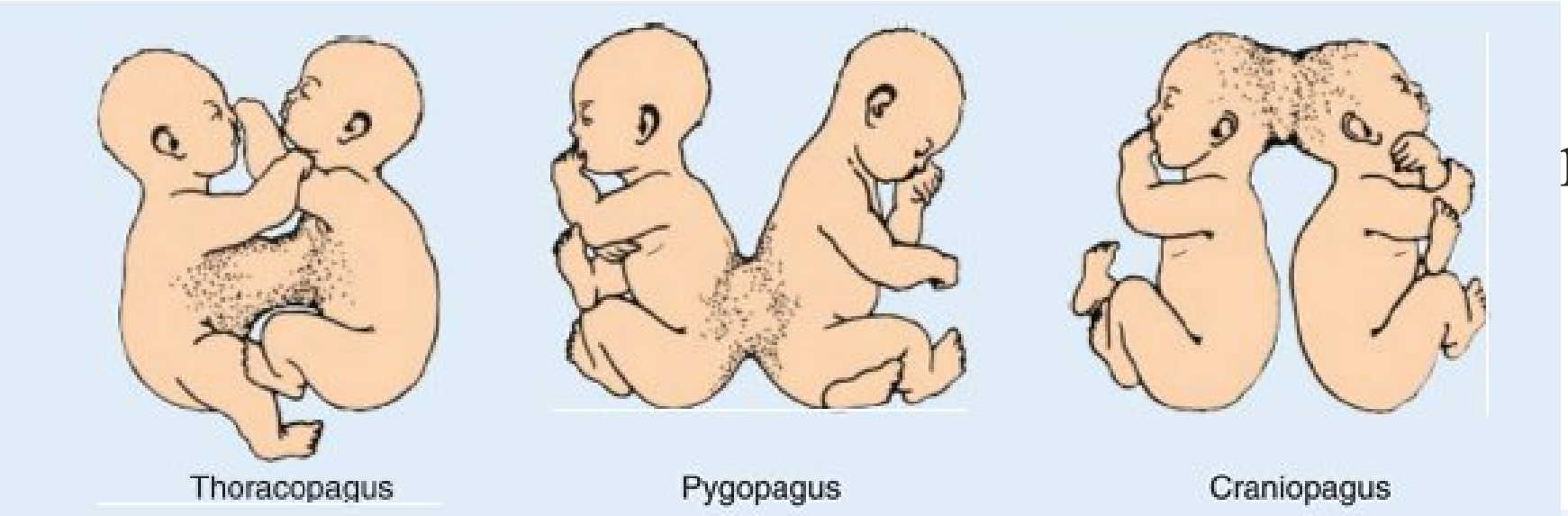
- Monochorionic twins share a placenta and therefore can have vascular connections between circulations.
- Feto–Fetal transfusion syndrome (FFTS): unbalanced blood flow through placental vascular anastomoses.
- Twin anemia–polycythemia sequence (TAPS): chronic unequal red-cell transfer.
- Intrauterine growth restriction (IUGR) can occur when placental sharing is unequal.
- MCMA twins are at risk of umbilical cord entanglement because both fetuses share one amniotic cavity

Feto–Fetal transfusion syndrome (FFTS)

- Occurs in monochorionic pregnancies with clinically significant unbalanced placental vascular connections.
- Donor twin → oligohydramnios and IUGR may develop.
- Recipient twin → polyhydramnios, polycythemia, heart failure, and hydrops in severe disease.
- Ultrasound assessment of chorion is most reliable early in pregnancy.

Conjoined twins

- Result from incomplete separation of a monozygotic embryonic disc, after day 13.
- The point of union determines the type: thoracopagus, omphalopagus, pygopagus, ischiopagus, or craniopagus.
- Shared organs may include the heart, liver, GIT, or nervous system
- Prenata my and shared



High-yield comparison

Feature	Dizygotic	Monozygotic
Origin	2 ova + 2 sperm	1 zygote → 2 embryonic units
Genetic similarity	Like siblings	Genetically identical
Sex	Same or different	Same
Typical chorionicity	Dichorionic	Dichorionic, or monochorionic
Typical amnionicity	Diamniotic	Diamniotic, or monoamniotic
Shared placenta	No true shared placenta	Possible
Risk of FFTS	No	Only in monochorionic twins

Remember: monozygotic does not automatically mean monochorionic.

Clinical approach

Step 1: Determine chorionicity and amnionicity by ultrasound.

Step 2: Establish fetal number, viability, growth, anatomy, and placental relationships.

Step 3: For monochorionic twins, monitor for FFTS, TAPS, IUGR, and other complications.

Step 4: For monoamniotic twins, monitor closely for cord-related complications.

Step 5: For suspected conjoined twins, define shared anatomy and refer to a specialized multidisciplinary center.